A MICROMACHINED SUSPENDED PLATE VISCOSITY SENSOR FEATURING IN-PLANE VIBRATIONS AND INTEGRATED PIEZORESISTIVE READOUT

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ABSTRACT

We present a novel resonant miniaturized sensor for the viscosity of liquids. The sensing element is a rectangular plate suspended by four beam springs. This plate vibrates in-plane. Thus, the plate's contribution to the device damping is low, but the plate increases the moving mass of the resonator. This principle aims at improving the quality factor, and gives additional freedom to the device designer. The sensor was fabricated by micromachining, and features Lorentz force actuation and piezoresistive readout. We present a device prototype and show by experiment, that the quality factor of the sensor is an appropriate measure for the viscosity.

KEYWORDS

Viscosity, density, suspended plate, MEMS, in-plane vibration

INTRODUCTION

In general, viscosity sensors based on micromachined structures, e.g., cantilevers or clamped-clamped beams, operate at lower frequencies and higher vibration amplitudes than thickness shear mode (TSM) resonators and surface acoustic wave devices [1–3]. Therefore, such sensors probe the low-frequency behavior of liquids, that is more comparable to the steady shear viscosity obtained by, e.g., conventional cone–plate viscometers. They are thus more suitable for the measurement of the viscosity of complex-structured or non-Newtonian liquids. We recently presented a sensor system based on a micromachined vibrating beam excited by Lorentz forces and featuring optical readout, and demonstrated its feasibility for nano-suspensions exhibiting a complex rheological behavior [4].

The fundamental mode of vibrating beams like in [1] exhibits a deflection in the out-of-plane direction. The flow field around the beam is rather complex with a significant damping due to compressional waves [5]. In contrast, sensors which mainly excite shear waves in the liquid, like the TSM resonator, feature much lower damping [3]. Furthermore, the highly sensitive optical readout method used in [1] is prone to vibrations of the measurement setup, does not allow obtaining the vibration amplitude of the device, and restricts the applicability of the sensor system to transparent liquids [4].

The novel suspended plate sensor aims at improving the

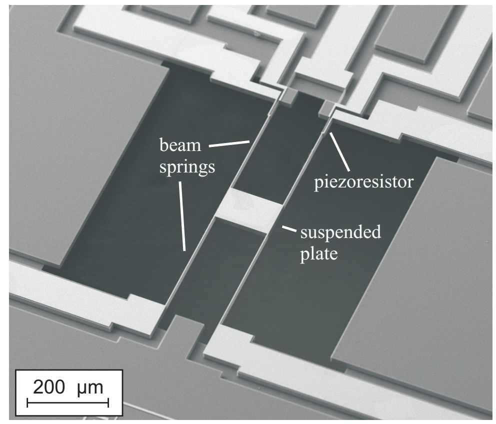

Figure 1: Scanning electron microscope (SEM) image of the suspended plate viscosity sensor.

quality factor of micromachined viscosity sensors. The concept of the device is a thin plate vibrating in the in-plane direction (Fig. 1). At the top and the bottom surface of the plate, shear waves are excited. These shear waves are characterized by a small penetration depth. Consequently, the contribution of the plate to the damping of the sensor vibration is low [6]. The dissipation losses of the resonator vibrating in liquid are mainly determined by the damping of the beam springs, and are, therefore, comparable to those of the clamped–clamped beam device [1]. However, the plate increases the moving mass of the sensor and, thus, the quality factor of the resonance.

The suspended plate structure is made from p-doped silicon, which is piezoresistive. The device utilizes this effect for the readout of the plate deflection. No additional fabrication steps to deposit piezoelectric or piezoresistive material or additional doping processes [7] are needed.

SENSOR DEVICE

Figure 2 depicts the schematic of the novel sensor device. The main element is a rectangular plate suspended by four beam springs. Through the conductive layer on the beams, a sinusoidal excitation current $i_\mathrm{e}(t)$ is driven. In the field of a permanent magnet (flux density $\vec{B} = -\vec{e}_z B_z$), Lorentz forces excite lateral vibrations of the plate and the springs. The ends of two beam springs (bottom of Fig. 2) are forked, forming two paths to carry the excitation currents, and the piezoresistive elements. Depending on the de-

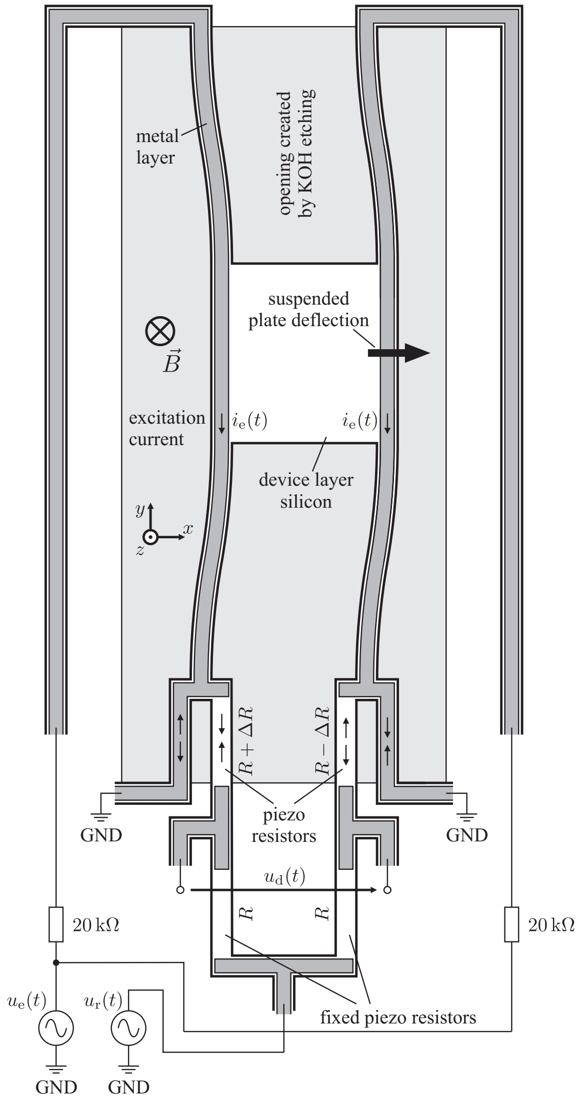

Figure 2: Schematic of the micromachined viscosity sensor. The in-plane movement of the plate is excited by the Lorentz forces arising from the excitation current $i_e(t)$ (sinusoidal, frequency f_e) and the magnetic flux density $\vec{B}$. The actual deflection is detected by a Wheatstone bridge of four piezoresistors (p-Si). Two of them are fixed, and two are mechanically stressed by the beam deflection. The Wheatstone bridge is fed with a sinusoidal voltage $u_r(t)$ at a frequency f_r. A lock-in amplifier locked to the difference frequency $f_r - f_e$ detects the beam deflection.

flection of the vibrating plate, those piezoresistors are subject to either compressive or tensile stress, their resistance changes accordingly. With two additional resistors, they form a Wheatstone bridge circuit driven by a readout voltage $u_r(t)$. The sensor readout yields a differential output voltage

$$u_d(t) = \frac{1}{2}\pi_l u_r(t)\sigma_l(t), \tag{1}$$

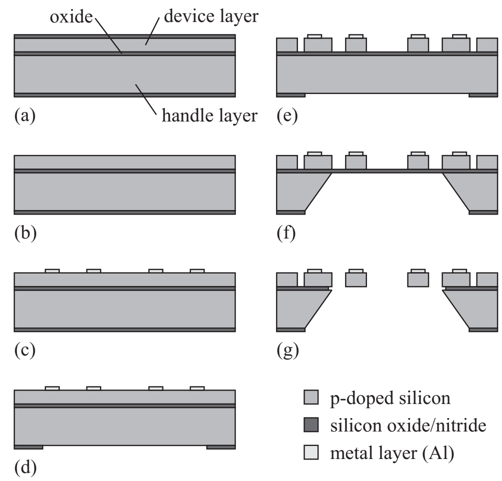

Figure 3: Simplified schematic of the fabrication process.

where $\pi_l = 7.18 \cdot 10^{-10}$ Pa^{-1} is the piezoresistive coefficient of the silicon layer [8], and $\sigma_l(t)$ is the mean mechanical stress acting in the piezoresistors. The sensor vibrates at the excitation frequency f_e, and therefore $\sigma_l(t) = \hat{\sigma}_l \cos(2\pi f_e t + \phi)$. Feeding the Wheatstone bridge at a different frequency f_r, $u_r(t) = \hat{u}_r \cos(2\pi f_r t)$, leads to an amplitude modulation in the piezoresistors, see equation (1). Therefore, the plate deflection can be obtained by setting a lock-in amplifier to either $|f_r - f_e|$ or $|f_r + f_e|$, and crosstalk from the excitation current to the sensor output is eliminated.

The sensor was fabricated on a silicon-on-insulator (SOI) wafer. The thicknesses of the device layer, the buried oxide layer, and the handle layer were 20 µm, 2 µm, and 350 µm, respectively. The device silicon layer was p-doped featuring a conductivity in the range from 0.11 Ωcm to 0.21 Ωcm. Both sides of the wafer were coated with a stack of silicon nitride and silicon oxide (Fig. 3a). First, the top side coating was removed by reactive ion etching (RIE, Fig. 3b). A 500 nm aluminium layer was vapor deposited and patterned using lift-off technique (Fig. 3c) to form the electrical connections. Subsequent annealing in vacuum was required to establish ohmic contacts between the metal layer and the silicon. Then openings were etched using RIE into the backside nitride-oxide stack, which were required as mask for the KOH etching process (Fig. 3d). The free-standing structure, the piezo resistors, and the conducting paths were formed by deep reactive ion etching (DRIE) on the wafer's front side (Fig. 3e). Subsequently, the wafer was KOH etched from the back-side (Fig. 3f), and the plate and the suspension springs were released by wet etching with buffered hydrofluoric acid (Fig. 3g). Figure 1 is a scanning electron microscope (SEM) image of the sensor device.

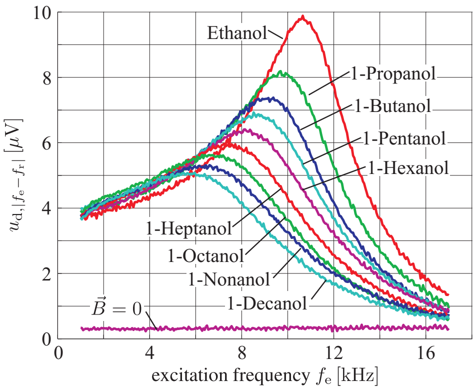

Figure 4: Resonance peaks of the suspended plate sensor immersed in a variety of nine sample liquids. The tenth result labeled $\vec{B} = 0$ was recorded with the electromagnet turned off, and indicates the noise floor.

EXPERIMENTAL RESULTS

The sensor used in the experiments featured a rectangular plate of $100\times100\times20\,\mu m^3$. Each of the four beam springs was 600 µm long, 5 µm wide, and 20 µm high. Two Agilent 33220A function generators provided the excitation current $i_e(t)$ and the readout voltage $u_r(t)$. In the excitation circuit, $20\,k\Omega$ series resistors were placed (Fig. 2). In these first experiments, we used small excitation currents, thus making sure not to destroy the device. The excitation current amplitude was 500 µA, and the readout voltage and frequency were $\hat{u}_r = 1$ V and $f_r = 99.4$ kHz, respectively. To provide the required magnetic field, we applied an electromagnet with a magnetic flux density of $B_z = 1.9$ T. Later experiments showed, that it is possible to increase the excitation current up to several mA, which allows using permanent magnets instead. A Stanford Research SR830 lock-in amplifier measured the output voltage of the bridge circuit, u_d. A third function generator, set to the differential frequency $|f_r - f_e|$, provided the reference signal for the lock-in amplifier. However, using this third signal source makes the measurement of the phase shift between the excitation current and the mechanical deflection impossible, as the mutual phase shifts of the function generators are arbitrary. A variety of nine sample liquids was used in the experiments. Their viscosities and densities are listed in Tab. 1.

The sensor was immersed in the sample liquids, and a frequency sweep measurement varying the excitation frequency f_e was taken for each liquid. Figure 4 depicts the sensor output voltage measured by the lock-in amplifier, $u_{d,|f_r-f_e|}(\omega_e)$ (rms), where $\omega_e = 2\pi f_e$. A tenth curve was recorded with the electromagnet turned off, and thus gives an idea of the noise floor. To extract the resonance frequency $f_0 = \omega_0/(2\pi)$ and the quality factor Q from the measurement results, the magnitude of a second order system,

$$u_{d,|f_r-f_e|}(\omega_e) = \left| \frac{A_0}{1 + j\omega_e \frac{1}{Q\omega_0} - \frac{\omega_e^2}{\omega_0^2}} \right|, \qquad (2)$$

was fit to the measurement data. The results are listed in Tab. 1. The resonance frequencies were in the range from 7.18 to 10.71 kHz, and the quality factors ranged from 1.2 to 2.75. Figure 5 depicts the relation between the damping factor $D = 1/(2Q)$ and the viscosity of the liquids η. The figure shows, that the damping of the sensor is dominated by the viscosity of the surrounding liquid and thus represents an appropriate measure for the viscosity. The relation between D and η is given by $D = 1.998\eta^{0.346}$.

Table 1: Dynamic viscosity and mass density of the sample liquids used in the experiments [9, 10], and the results of curve-fitting a second order system, equation (2), to the measurement data of Fig. 4: f_0 is the resonance frequency, Q is the quality factor, and $D = 1/(2Q)$ is the damping factor.

Liquid	η [mPa·s]	ρ [kg/m^3]	f_0 [kHz]	D	Q
ethanol	1.06	789	10.71	0.18	2.75
1-propanol	1.90	804	9.92	0.23	2.17
1-butanol	2.52	810	9.38	0.26	1.92
1-pentanol	3.44	814	9.11	0.29	1.75
1-hexanol	4.49	814	8.82	0.30	1.64
1-heptanol	5.94	822	8.24	0.33	1.50
1-octanol	7.37	827	7.76	0.36	1.36
1-nonanol	9.10	827	7.64	0.39	1.27
1-decanol	10.97	830	7.18	0.42	1.20

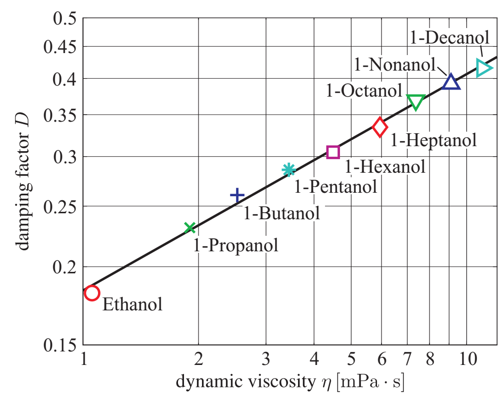

Figure 5: Damping factor D versus viscosity η. The markers indicate the measurement results, whereas the solid line depicts the fit result, $D = 1.998\eta^{0.346}$.

CONCLUSIONS

The presented viscosity sensor is based on a rectangular plate featuring an in-plane mode of vibration. The feasibility of the novel viscosity sensor was demonstrated by experiments. The integrated piezoresistive readout was successfully used to measure the frequency characteristics of the plate immersed in liquid. The sensor output voltages were in the range of up to 10 µV with a noise floor at 350 nV.

At the top and bottom faces of the rectangular plate, mainly shear waves are excited. The contribution of the plate to the damping of the device is therefore low [11]. Therefore, increasing the size of the plate only results in a small increase of the damping, whereas the higher moving mass enhances the quality factor of the device immersed in liquid. Higher quality factors lead to increased deflection amplitudes and better signal-to-noise ratios of the readout. The experiments with a first device resulted in quality factors in the range from 1.2 to 2.75. Such quality factors were previously achieved by clamped–clamped beam sensors [1, 2]. However, the suspended plate sensor proved mechanically stable. Hence, devices featuring larger plates can be fabricated. A theoretical model shows, that such devices feature quality factors of 10 and above [11].

Varying the size of the rectangular plate also allows the design of devices that exhibit different resonance frequencies. An array of such devices on a single chip would probe the viscosity of liquids at several frequencies simultaneously, which could be a promising approach to measure the behaviour of viscoelastic fluids.

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REFERENCES

[1] C. Riesch, E. K. Reichel, A. Jachimowicz, F. Keplinger, and B. Jakoby, "A novel sensor system for liquid properties based on a micromachined beam and a low-cost optical readout," in *IEEE Sensors Conf.*, Atlanta, USA, Oct. 28–31, 2007, pp. 872–875.

[2] I. Etchart, H. Chen, P. Dryden, J. Jundt, C. Harrison, K. Hsu, F. Marty, and B. Mercier, "MEMS sensors for density-viscosity sensing in a low-flow microfluidic environment," *Sens. Actuators A*, vol. 141, pp. 266–275, 2008.

[3] K. K. Kanazawa and J. G. Gordon II, "Frequency of a quartz microbalance in contact with a liquid," *Anal. Chem.*, vol. 57, pp. 1770–1771, 1985.

[4] C. Riesch, E. K. Reichel, A. Jachimowicz, F. Keplinger, and B. Jakoby, "A micromachined doubly-clamped beam rheometer for the measurement of viscosity and concentration of silicon-dioxide-in-water suspensions," in *IEEE Sensors Conf.*, Lecce, Italy, Oct. 26–29, 2008, pp. 391–394.

[5] B. Weiss, E. K. Reichel, and B. Jakoby, "Modeling of a clamped-clamped beam vibrating in a fluid for viscosity and density sensing regarding compressibility," *Sens. Actuators A*, vol. 143, pp. 293–301, 2008.

[6] J. H. Seo and O. Brand, "High Q-factor in-plane-mode resonant microsensor platform for gaseous/liquid environment," *J. Microelectromech. Syst.*, vol. 17, pp. 483–493, 2008.

[7] E. J. Eklund and A. M. Shkel, "Single-mask fabrication of high-G piezoresistive accelerometers with extended temperature range," *J. Micromech. Microeng.*, vol. 17, pp. 730–736, 2007.

[8] Y. Kanda, "Piezoresistance effect of silicon," *Sens. Actuators A*, vol. 28, pp. 83–91, 1991.

[9] D. R. Lide, Ed., *CRC Handbook of Chemistry and Physics*, 83rd ed. CRC Press, 2002.

[10] D. S. Viswanath, T. K. Ghosh, D. H. L. Prasad, N. V. K. Dutt, and K. Y. Rani, *Viscosity of Liquids*. Springer, 2007.

[11] C. Riesch, E. K. Reichel, A. Jachimowicz, J. Schalko, P. Hudek, B. Jakoby, and F. Keplinger, "A suspended plate viscosity sensor featuring in-plane vibration and piezoresistive readout," in preparation.

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